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June

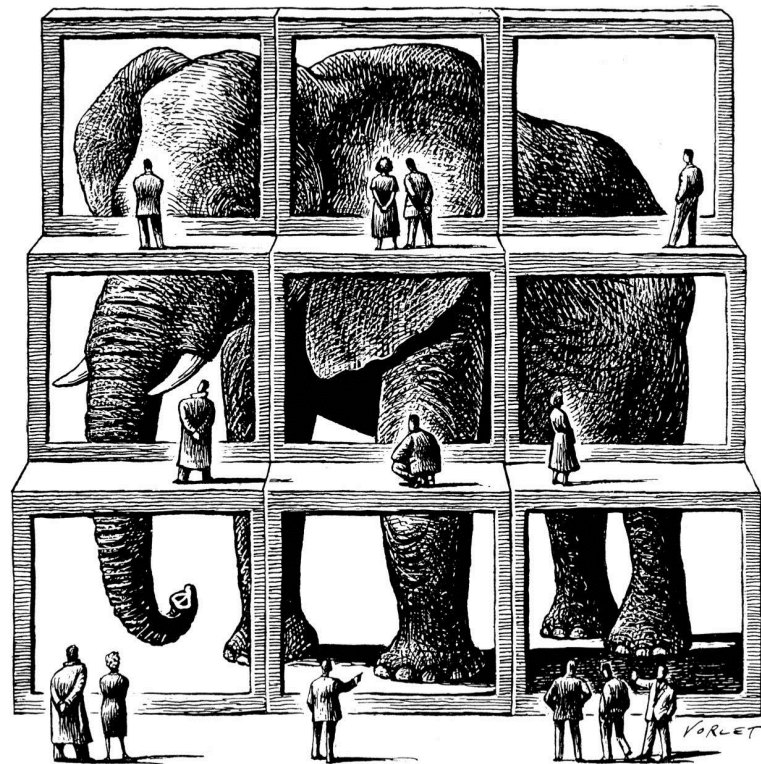
**MASSIVE
TRANSFUSION**
with
FREQUENT DIGRESSIONS

ICU Regional Teaching

Jake Barlow

Overview

1. Context
2. Transfusion Medicine
3. Haemostatic Resuscitation
4. Massive Transfusion



Context

- Haemostatic resuscitation
- Theory
 - Designed for trauma
 - Conducted by US military, or in North American
- Practice
 - Mostly GI/obstetric/surgical
 - If trauma, mostly blunt

Definitions

- Massive Transfusion
 - Whole blood volume in 24 hours
 - 10 PRBC in 24 hours
 - $\geq 50\%$ of blood volume in 4 hours
 - $>40\text{mL/kg}$ in children.

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Definitions

- Critical bleeding
 - Life-threatening haemorrhage likely to result in massive transfusion
 - Bleeding in a critical area (e.g. intracranial, intraocular, intraspinal), resulting in morbidity or mortality



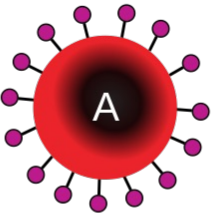
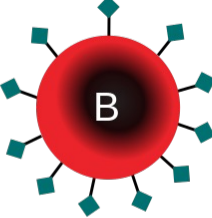
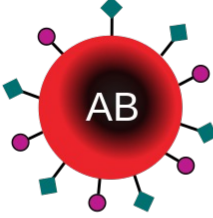
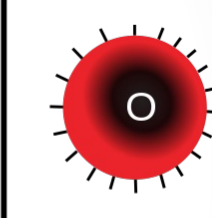
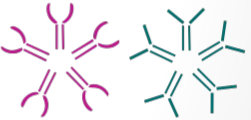


A. E. Elias
1st Div.

Transfusion

- Transfusion reactions
 - Blood types
 - Reactions
- (Fractionated) blood products

A Brief Refresher on Blood Types

	Group A	Group B	Group AB	Group O
Red blood cell type				
Antibodies in plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in red blood cell	 A antigen	 B antigen	 A and B antigens	None

Transfusion Reactions

- Adverse event from transfusion of blood
- Acute/Delayed
- Immunological/Non-immunological

Acute + Immunological

Reaction	Incidence	Mechanism
ABO Mismatch	1:40,000	ABO incompatibility causing rapid intravascular haemolysis, which may cause chest pain, jaundice, shock, and DIC . RhD-reactions tend to cause extravascular haemolysis.
Haemolytic (acute)	1:76,000 (1:1.8 million fatal)	Immunological destruction of transfused cells (Type II hypersensitivity). Presents with fever, tachycardia, pain, progressing to distributive shock
Febrile, non-haemolytic	~1:100	Cytokine release from stored cells causing a mild inflammatory reaction, with temperature rising to $\geq 38^{\circ}\text{C}$ or $\geq 1^{\circ}\text{C}$ above baseline (if $>37^{\circ}\text{C}$). Benign - but requires exclusion of a haemolytic reaction.
Urticaria	1:100	Hypersensitivity to plasma proteins in the transfused unit
Anaphylaxis	1:20,000	Type I hypersensitivity reaction to plasma protein in transfused unit
TRALI	Variable	Donor plasma HLA activates recipient pulmonary neutrophils, causing fever, shock, and non-cardiogenic pulmonary oedema

Delayed + Immunological

Reaction	Incidence	Mechanism
Delayed haemolytic transfusion reaction	1:2,500	Development of sensitisation with the reaction occurring 2-14 days after a single exposure. Typically Kidd, Duffy, Kell antibodies.
Post-transfusion Purpura	Rare	Alloimmunisation to Human Platelet Antigen causing sudden self-limiting thrombocytopenia
TA-GVHD	Rare	Transfused lymphocytes recognise host HLA as positive causing marrow aplasia, with mortality >90%
Alloimmunisation	1:100 (RBC antigens), 1:10 (HLA antigens)	Previous sensitisation leading to antibody production on re-exposure.
Transfusion-related Immune Modulation	Not known	Transient immunosuppression following transfusion potentially due to cytokine release from leukocytes

Acute + Non-Immunological

Reaction	Incidence	Mechanism
Non-immune mediated haemolysis	Rare	Due to physicochemical damage to RBCs (freezing, device malfunction). May lead to haemoglobinuria, haemoglobinaemia, tachycardia and fevers.
Sepsis	1:75,000 (platelets), 1:500,000 (RBC)	Contamination during collection or processing. Most common organisms are those which use iron as a nutrient and reproduce at low temperatures, e.g. <i>Yersinia Pestis</i> .
Transfusion Related Circulatory Overload (TACO)	< 1:100	Rapid increase in intravascular volume in patients with poor circulatory compliance or chronic anaemia. May result in pulmonary oedema and be confused with TRALI.

Delayed + Non-Immunological

Reaction	Incidence	Mechanism
Iron Overload	Chelation after 10-20 units, organ dysfunction 50-100 units	Each unit of PRBC contains ~250mg of iron, whilst average excretion is $1\text{mg}\cdot\text{day}^{-1}$.

Blood

Products

- **Packed cells**
- **Fresh frozen plasma**
- **Cryoprecipitate**
- **Platelets**
- **Whole blood**

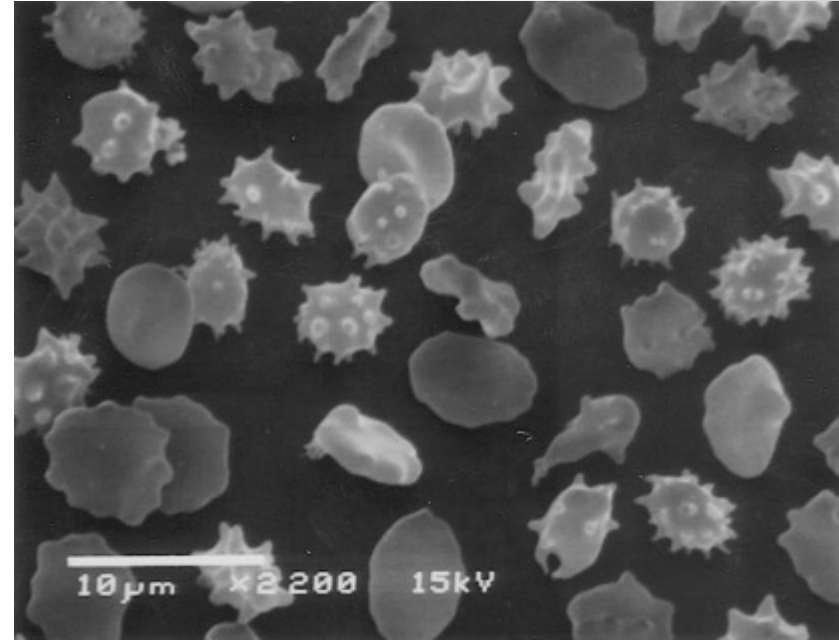
- Factor concentrates
 - Factor VIII
 - Factor IX
 - **PCC** (II, IX, X, +/- VII)
 - FEIBA
 - Factor XIII
 - AT III
 - Protein C
- Albumin
- Immunoglobulins
 - Hep B/Tetanus/VZV/Rabies/HSV
 - Anti-D
 - IVIG
- Granulocytes
- Other
 - C1-esterase inhibitors

Packed Cells

- Cytopheresed donor blood
- 1 unit $\approx$ 200-300mL
- Raises Hb by $\sim$ 10g/dL
- Leukoreduced
- Minimal plasma

Storage Lesion

- ↓ 2-3 DPG
- ↓ Deformability
- ↑ Fragility



Packed Cells Storage

- **Begin once removed** from refrigerator
- **Complete** within **4 hours**
- **Returned** to controlled storage **within 30 minutes** of removal
- Cannot be returned if out of controlled storage for >30 minutes

Fresh Frozen Plasma

- Plasma remaining after cytappheresis
- Frozen and **thawed** prior
- 1 unit $\approx$ 200-300 ml
- Dose **10-15ml/kg**
- ABO compatibility *desirable*

Fresh Frozen Plasma

- **Commence** within **30 minutes**
- **Complete** within **4 hours** of thawing
- Cannot be returned after 30 minutes
- Cannot be re-frozen
 - Can be refrigerated for up to 24 hours at 2-6°C and re-issued

Cryoprecipitate

- FFP derivative
- Contains concentrated:
 - **Fibrinogen**
 - Factor VIII
 - Factor XIII
 - Fibronectin
 - vWF

Cryoprecipitate

- **Commence** within **30 minutes**
- **Complete** within **4 hours**
- Cannot be **returned** after **30 minutes** out of controlled storage

Platelets

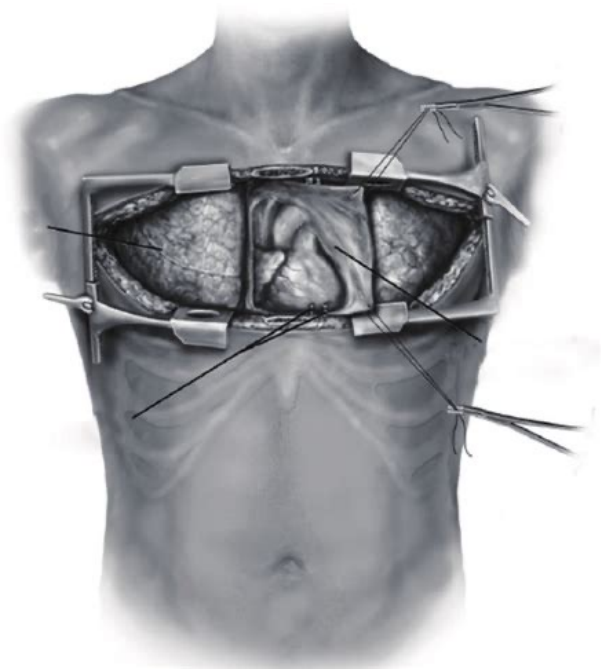
- 1 unit $\approx$ 50mL
- $\uparrow$ by $\sim 10 \times 10^9$ /L.
- Usually come in 4-unit bags

Platelets

- **Commence** as soon as received
- **Completed** within **1 hour** of issue
- *May* be able to be returned after 1 hour, conditional on storage

Whole Blood

- Full haemostatic function
- ↑ Complexity of type matching
 - Group-specific
 - Low-titre type O
- Fresh WB
- Stored WB



Predicting Massive Transfusion



Assessment of Blood Consumption

- HR ≥ 120
- SBP ≤ 90
- Penetrating injury
- Positive FAST scan



Transfusion Medicine Handbook Third Edition, 2016



A Guide to the Clinical Use
of Blood Components,
Blood Products and Blood
Transfusion Procedures in
New Zealand

Protocol Activation Criteria

The massive transfusion protocol may be activated for a specific patient when the following conditions are met:

- There is massive bleeding with either shock or abnormal coagulation
- The patient has been assessed as requiring the protocol by an experienced clinician



- The patient is transfused 3 units of RBC or whole blood (either type-specific or emergency O RhD negative)
- The patient is reassessed and, if criteria are met, the MTP is activated



- The clinician (or delegate) must notify Blood Bank of the activation, informing them of the patient's name, NHI and clinical area



Injury Patterns

- Uncontrolled:
 - Truncal bleeding
 - Junctional haemorrhage
 - Large soft tissue bleeding
- Amputations and mangled extremities
- Clinical coagulopathy
- Severe hypothermia

Haemostatic Resuscitation

- Permissive hypotension
- Balanced transfusion
- Avoid crystalloid
- Normothermia
- Correct acidosis
- Calcium
- TXA
- Correct coagulopathy

Principles of Massive Transfusion

- Access
- Clinical situation outpaces investigations
- Empiric treatment of coagulation

IV Access





Investigations

- **Blood group**
- ABG
 - pH
 - Ca
 - Hb
- Coagulation
 - APTT
 - PT/INR
 - Fibrinogen
- Viscoelastic

Typing in Massive Transfusion

- Typing balances:
 - Risk of haemolysis
 - Urgency of transfusion
 - Allocation of scarce resources

**SAVE LIVES
GIVE BLOOD**

Blood types are divided into 8 main groups.

- 38% are O+
- 9% are O-
- 32% are A+
- 6% are A-
- 9% are B+
- 2% are B-
- 3% are AB+
- 1% are AB-

Blood groups and Families
A child's blood group is linked to their parents.
Below are combinations of parents' blood types, and possible combinations of their biological children.
Note: there are some rare exceptions.

ABO Blood Groups

Parents	Possible Children
O & O	O
O & A	O or A
O & B	O or B
O & AB	A or B
A & A	O or A
A & B	O, A, B or AB
A & AB	A, B or AB
B & B	O or B
B & AB	A, B or AB
AB & AB	A, B or AB

Rh(D) Groups

Parents	Children
+ & +	+ & -
+ & -	+ & -
- & -	-

Source: NZBS Donor Database, September 2013

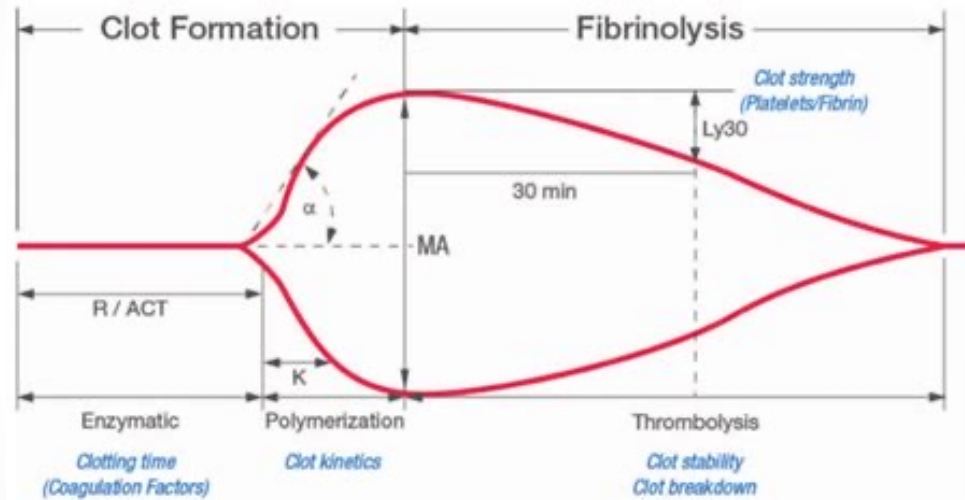
Blood Typing

1. Blood Group
ABO/Rh. ~15 minutes.
2. Minor antibody screen
3. Crossmatch
 - ~5 minutes at best
 - Hours at worst

Product Allocation

- Untyped
- Group-specified uncrossmatched
- Group-specific partial crossmatch
- Group-specific fully crossmatched

Investigations



Code Crimson MHP
Trauma + ABC Score ≥ 2
+ senior clinician approval

Standard MHP
Medical or Surgical Bleeding

Obstetric MHP

2g Tranexamic Acid

1g Tranexamic Acid

1g Tranexamic Acid

Send Group + Screen

Initiate: Call Blood Bank 24015, Provide Patient Details

State "I am requesting (Crimson, Standard, Obstetric) Stat Pack"

Crimson Stat Pack
2 RBC & 2 FFP or 2 WB

Standard Stat Pack
2 RBC or 2 WB

Obstetric Stat Pack
2 RBC

Reassess: Ongoing Massive Bleeding + Shock?

Activate MHP, Identify Transfusion Coordinator, Call Blood Bank 24015

State "I am activating (Crimson, Standard, Obstetric) MHP"

Crimson Pack 1
Straight to Pack 2

Standard Pack 1
2 RBC & 2 FFP or 2 WB

Obstetric Pack 1
2 RBC, 3 Cryo

Pack 2
4 RBC, 4 FFP
3 Cryo

1g Calcium with every pack

Pack 3
4 RBC, 4 FFP
1 Platelet*

Alternating
packs 2 & 3 until
bleeding slowed

Then stop MHP,
and start targeted
transfusion

Bloods:

- repeat every 30min
- Blood gas
- iCa²⁺
- FBC
- Coags
- Fibrinogen
- Viscoelastic if available e.g. TEG®

Endpoints

- No evidence of further haemorrhage
- C
 - SBP 80-90mmHg
 - MAP >50mmHg
- E
 - Temperature >35°C
- F
 - pH >7.3
 - iCa²⁺ >1.1mmol/L
- H
 - Hb >70g/L
 - INR <1.5
 - APTT <40s
 - Fibrinogen >2g/dL
 - Platelet >50-100×10⁹/L

Complications

Blood:

- Citrate toxicity
- Acidosis
- Hyperkalaemia
- Hypokalaemia
- Hyperbilirubinaemia

Procedural:

- Air embolism
- Hypothermia

The End

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